

ABDULLAH HASHMAT

AI/ML Researcher • LLMs & Agentic Systems • NLP & Computational Biology
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EDUCATION

BS Computer Science | [Lahore University of Management Sciences \(LUMS\)](#) Aug 2021 – May 2025

Relevant Coursework: CS6304 Advanced Machine Learning, CS5302 Speech Processing with Generative AI, CS6303 Large Language Models, CS331 Artificial Intelligence, CS487 Cloud Development, CS437 Deep Learning

TECHNICAL SKILLS

Programming: Python, C++, C, C#, Java, SQL, Haskell

AI/ML: PyTorch, Transformers, LLMs, Agentic AI, RAG, NLP, Computer Vision, Fine-tuning, Knowledge Distillation

Cloud & MLOps: AWS (Lambda, ECS, DynamoDB, S3), Docker, Distributed Training, Model Deployment, Serverless Architecture

Frameworks & Tools: Hugging Face, LangChain, RDKit, Git, Linux, React, MERN, BERT, RoBERTa, DeBERTa

EXPERIENCE

Research Fellow | [Max Planck Institute for Software Systems \(MPI-SWS\)](#) | Germany, KL Apr 2026 – Present

Supervised by Dr. Abhilasha Ravichander

- Designing an LLM-driven agentic system for inverse molecular design, combining generative models with structured tool-calling (RDKit, docking, QSAR) for multi-objective molecular optimization.
- Building closed-loop generation pipelines with iterative refinement and feedback-driven reasoning, improving validity, diversity, and binding affinity across candidate molecules.

Researcher | [Center for Speech and Language Technologies \(CSaLT\), LUMS](#) | Lahore, PK (Remote) Sep 2024 – Present

Supervised by Dr. Agha Ali Raza & Dr. Usman Naseem

- Developing self-evolving agentic frameworks with built-in cultural awareness, enhancing LLM reasoning and tool-calling performance while reducing Western-centric alignment biases.
- Published PakBBQ, Pakistan's first locally contextualized QA bias benchmark (17,180 QA pairs across 8 socio-cultural categories); accepted at EMNLP 2025 Main Conference.
- Led development of Pak3H, a human-validated Urdu benchmark suite (28,000 samples) evaluating LLM alignment across helpfulness, harmlessness, and honesty; exposing significant cross-lingual degradation.

Lead Researcher | [German Research Center for Artificial Intelligence \(DFKI\) & RPTU](#) | Germany, KL Jul 2025 – Mar 2026

Supervised by Dr. Nabeel Asim

- Developed a multi-representation SMILES-based Molecular Property Prediction model using a multi-granular transformer framework leveraging atom-, functional-group-, and branch-level representations.
- Pre-trained BERT, RoBERTa, DeBERTa, and state-space models on large-scale SMILES corpora; distilled into a unified student via MLM + FitNet distillation with dynamic gating.
- Achieved state-of-the-art performance across all 22 ADMET benchmark tasks, outperforming models that only excel on task subsets.
- Built and processed a 500M+ compound dataset spanning 150+ molecular properties using large-scale data aggregation, web scraping, and distributed GPU clusters.

MLOps Engineer | [Darcheeni \(AI Healthcare Startup, LUMS\)](#) | Lahore, PK May 2024 – May 2025

- Led backend development for an AI-driven clinical assistant integrating ASR, RL-based decision models, and LLM-driven response generation for real-time patient triage.
- Optimized speech recognition pipelines through data processing, model fine-tuning, and LLM-based post-processing, improving transcription quality across diverse clinical settings.
- Designed and deployed serverless infrastructure on AWS (Lambda, DynamoDB, Auth0) with autoscaling for scalable, low-latency model inference.

Primary Researcher | [Center of Urban Informatics, Technology and Policy \(CITI\)](#) | Lahore, PK Sep 2024 – Mar 2025

Supervised by Dr. Muhammad Tahir

- Proposed a confidence-aware multi-teacher knowledge distillation framework to mitigate texture/shape inductive biases and improve OOD generalization in medical image classification.

- Ensembled shape- and texture-biased ViT and VGG16 teachers using entropy-based logit weighting; trained ViT-Small student via KL divergence with dynamic bias propagation.
- Achieved balanced cue reliance (shape 0.49 / texture 0.51) and outperformed single-teacher and naive KD baselines across ImageNet-1k, Stylized-ImageNet, and OOD benchmarks.

Sales Performance Intern
|
 Delivery Hero (Foodpanda)
 |
 Lahore, PK

Jun 2024 – Aug 2024

- Conducted large-scale EDA on vendor onboarding and QC data to identify bottlenecks, reducing QC rejection rate by 15%.
- Developed an agentic Python workflow using Google APIs to automate daily data pulls, invoice generation, and vendor distribution via WhatsApp, reducing process time by 35%.

Data Science Intern
|
 Pakistan Television Network (PTV)
 |
 Lahore, PK

Jun 2023 – Jul 2023

- Revamped ticker headline system using ML text similarity models, achieving a 20% reduction in headline redundancy.
- Built a predictive content analysis model that improved viewer engagement metrics by 10% through data-driven strategies.

PUBLICATIONS & RESEARCH

PakBBQ: Culturally Adapted Bias Benchmark for QA
—
 EMNLP 2025 Main Conference (2025)
 [Paper Link](#)

Pak3H: Evaluating the Cost of Cultural Mismatch in LLM Alignment
—
 Under Review (2026)
 [Link](#)

TEACHING ASSISTANTSHIPS

AI600 Machine Learning (Masters in AI Program)
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 LUMS

Jan 2025 – May 2025

CS5302 Foundations of Generative AI
 |
 LUMS

Jan 2025 – May 2025

CS535 Machine Learning
 |
 LUMS

Sep 2024 – Dec 2024

CS100 Computational Problem Solving
 |
 LUMS

Sep 2023 – Dec 2023

SELECTED PROJECTS

AI-Driven Stock Insights Platform
(AWS + Claude API)
[Link](#)

Sep 2024 – Dec 2024

- Built and deployed full-stack real-time stock analysis platform using AWS Lambda, S3, ECS, and Neon Postgres; integrated Claude-based chatbot for natural language investment queries.
- Achieved <200ms latency with auto-scaling to millions of users via serverless architecture and ECS Fargate.

Bias Mitigation in Deep Learning Vision Models
[Link](#)

Sep 2024 – Dec 2024

- Implemented multi-teacher single-student knowledge distillation (ViT + VGG16) to address inductive biases; achieved 12% bias reduction on cue-conflict datasets under distributional shift.

Agentic Data Science Chatbot
[Link](#)

Jan 2024 – May 2024

- Built an agentic AI chatbot for automated EDA, statistical analysis, and data cleaning across 10+ formats; fine-tuned Llama 3 and deployed on Hugging Face via Gradio.

LEADERSHIP & ACTIVITIES

TechStars Startup Weekend Lahore — Convenor

Jul 2022 – Mar 2023

- Organized Startup Weekend Lahore 2023 with National Incubation Center & Google for Startups; managed 40+ startups, 150+ attendees, team of 60, and secured 800K+ PKR in sponsorships.

SPADES (PSIFI) — Assistant Director, Socials & Operations

Sept 2022 – May 2023

- Led organisation of PSIFI's flagship event, coordinating social functions and concerts for 1,400+ attendees